

Predictive N_T — Six-Component COSMO-SAC-2010

RESEARCH DIAGNOSTIC — NOT ACCEPTED — NOT RELEASE ELIGIBLE

Executive summary

Project 209 completed 2 frozen candidate stage-count trials. A converged process-sensitivity trial is not an accepted Predictive N_T unless the separate scientific qualification is satisfied.

Project reference	209
Job status	COMPLETED — report generated automatically
Trials completed	2 / 2
Accepted Predictive N_T	NOT ASSIGNED
Established theoretical stages	Not established
Pilot validated	No
Calibration required	Yes
Release eligible	No

ENGINEERING RESULT

Preferred engineering candidate: NOT ASSIGNED

No governed preferred-candidate selection rule is persisted for this run. The report therefore does not invent or hardcode a stage selection. Accepted Predictive N_T remains a separate scientific qualification result.

COSMO-SAC sulfur prediction: NOT CALCULABLE

1. Frozen Design Basis

OWNER-CONTROLLED / SAVED STAGE-1 INPUT AUTHORITY

Project reference	209
RRBO grade	SN300
Operating temperature	50 °C
Operating pressure	2.0 bar(g)
Phase configuration	nmp-continuous-rrbo-dispersed
Solvent/oil mass ratio	1.5
Maximum stages	2
SAT identity	n-dodecane
MONO identity	n-propylbenzene

Feed composition — mass basis

SAT	MONO	DI	POLY	PA	NMP
65 wt%	20 wt%	5 wt%	3 wt%	5 wt%	2 wt%

Saved product targets

Maximum total aromatics	10 wt% NMP-free
Maximum polar aromatics	0.5 wt% NMP-free
Minimum saturates	90 wt% NMP-free
Minimum RRBO recovery	90 wt%
Maximum NMP in raffinate	1 wt% full stream
Raffinate sulfur target	1000 ppm (pre-pilot estimate only)

1. Frozen Design Basis — Sulfur Allocation

PRE-PILOT ASSUMPTION — saved with the completed Stage-1 snapshot

Component	Feed wt%	Sulfur allocation %	Feed-basis sulfur contribution ppm
SAT	65	0	0
MONO	20	20	700
DI	5	30	1050
POLY	3	40	1400
PA	5	10	350
Feed sulfur		3500 ppm	
Raffinate sulfur target		1000 ppm	
Allocation total		100 %	

This allocation is an owner-controlled pre-pilot engineering assumption. It is not a hidden report constant and it does not make sulfur a COSMO-SAC component.

2. Overall N_T Engineering Comparison

Primary process comparison. Product composition is wt% on an NMP-free hydrocarbon basis; NMP is wt% of total raffinate.

N_T	Solver	Recovery %	NMP %	SAT %	Arom %	PA %	SAT loss %	MONO rem %	DI rem %	POLY rem %	PA rem %	Sulfur ppm	Target	Scientific
1	CLOSED	95	—	—	10	1	—	50	62.5	66.67	75	N/C	PASS	NOT QUALIFIED
2	CLOSED	95	—	—	10	1	—	50	62.5	66.67	75	N/C	PASS	NOT QUALIFIED

Recovery = NMP-free RRBO recovery. Removal/loss = component-relative boundary removal. Sulfur = PRE-PILOT ALLOCATION ESTIMATE ONLY. Unconverged values are diagnostic only and receive no engineering PASS/FAIL.

3. Component Extraction Performance

Engineering trade-off between RRBO recovery and aromatic/polar-component removal

N_T	Solver Status	RRBO Recovery %	SAT Loss %	MONO Removal %	DI Removal %	POLY Removal %	PA Removal %
1	CLOSED	95	—	50	62.5	66.67	75
2	CLOSED	95	—	50	62.5	66.67	75

SAT is reported as SAT LOSS. NMP removal/extraction percentage is NOT APPLICABLE.

4. Pre-Pilot Sulfur Allocation Estimate

ESTIMATE ONLY — NOT A COSMO-SAC SULFUR PREDICTION — NOT PILOT VALIDATED

N_T	Solver	SAT ppm	MONO ppm	DI ppm	POLY ppm	PA ppm	Total ppm	Target ppm	Estimate Target Status
1	CLOSED	—	—	—	—	—	NOT CALCULABLE: MISSING_PERSISTED_RECOVERY_OR_COMPONENT_REMOVAL	1000	NOT_CALCULABLE
2	CLOSED	—	—	—	—	—	NOT CALCULABLE: MISSING_PERSISTED_RECOVERY_OR_COMPONENT_REMOVAL	1000	NOT_CALCULABLE

Formal COSMO-SAC sulfur prediction: NOT CALCULABLE

5. Engineering Stage Selection

Preferred engineering candidate, accepted Predictive N_T, and established theoretical stages are separate states

Preferred engineering/process-sensitivity candidate	NOT ASSIGNED
Accepted Predictive N_T	NOT ASSIGNED
Established theoretical stages	NOT ESTABLISHED

SELECTION GOVERNANCE

No explicit deterministic preferred-candidate selection rule is persisted for this completed run. No stage is selected by this report. Unconverged trials and scientifically unqualified trials can never become an accepted Predictive N_T through presentation logic.

6. Scientific Qualification Summary

Exact persisted evidence; PASS is never inferred from the absence of a blocker

Six-component molecular basis	PERSISTED
Stage-1 authority	PERSISTED
Primary solver closure	PASS
Secondary / multistart closure	PASS
Global TPD stability	BLOCKED_FAIL_CLOSED
Sulfur thermodynamic prediction	NOT CALCULABLE
Pilot validation	NOT VALIDATED
Calibration requirement	REQUIRED
Release eligibility	NOT ELIGIBLE
Accepted Predictive N_T	NOT ASSIGNED

Main-report blocker summary

No trial blocker codes persisted.

Appendix A — N_T=1 Numerical Diagnostics

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates PASS · maximum component-balance residual 0.000e+0

Primary termination GTOL · Residual closure CLOSED · Maximum scaled equation residual 0.000e+0 · Branch comparison EVALUATED (0.000e+0)

Blockers: None

Product targets — MASS BASIS (NMP-free hydrocarbon basis except total-raffinate NMP wt%)

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	—	—	—
maximum Nmp Raffinate Wt	—	—	—
minimum Raffinate Saturates Wt	—	—	—
maximum Raffinate Total Aromatics Wt	—	—	—
maximum Raffinate Polar Aromatics Wt	—	—	—
target Raffinate Sulfur Ppm	—	—	—

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP
Removal / loss	3.3333	50	62.5	66.67	75	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP
oil Feed	6.0000e-1	1.0000e-1	8.0000e-2	6.0000e-2	4.0000e-2	1.0000e-2
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	1.0000e+0
final Raffinate	5.8000e-1	5.0000e-2	3.0000e-2	2.0000e-2	1.0000e-2	2.0000e-2
final Extract	2.0000e-2	5.0000e-2	5.0000e-2	4.0000e-2	3.0000e-2	9.9000e-1

Overall six-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	0.000e+0	0.000e+0	0.000e+0	0.000e+0	0.000e+0	0.000e+0

Appendix C.1 — N_T=1 Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	0.000e+0	PASS	0.000e+0	1.000e+0	1.000e+0	0.000e+0	0.000e+0

Appendix B — N_T=1 Six-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
1 R	0.58	0.05	0.03	0.02	0.01	0.02
1 E	0.02	0.05	0.05	0.04	0.03	0.99

Appendix A — N_T=2 Numerical Diagnostics

RESEARCH DIAGNOSTIC — NOT ACCEPTED · numerical gates PASS · maximum component-balance residual 0.000e+0

Primary termination GTOL · Residual closure CLOSED · Maximum scaled equation residual 0.000e+0 · Branch comparison EVALUATED (0.000e+0)

Blockers: None

Product targets — MASS BASIS (NMP-free hydrocarbon basis except total-raffinate NMP wt%)

Target	Status	Limit	Calculated
minimum Nmp Free Recovery Pct	—	—	—
maximum Nmp Raffinate Wt	—	—	—
minimum Raffinate Saturates Wt	—	—	—
maximum Raffinate Total Aromatics Wt	—	—	—
maximum Raffinate Polar Aromatics Wt	—	—	—
target Raffinate Sulfur Ppm	—	—	—

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP
Removal / loss	3.3333	50	62.5	66.67	75	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP
oil Feed	6.0000e-1	1.0000e-1	8.0000e-2	6.0000e-2	4.0000e-2	1.0000e-2
fresh Nmp	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	1.0000e+0
final Raffinate	5.8000e-1	5.0000e-2	3.0000e-2	2.0000e-2	1.0000e-2	2.0000e-2
final Extract	2.0000e-2	5.0000e-2	5.0000e-2	4.0000e-2	3.0000e-2	9.9000e-1

Overall six-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP
Residual	0.000e+0	0.000e+0	0.000e+0	0.000e+0	0.000e+0	0.000e+0

Appendix C.1 — N_T=2 Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
2	0.000e+0	PASS	0.000e+0	1.000e+0	1.000e+0	0.000e+0	0.000e+0

Appendix B — N_T=2 Six-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP
2 R	0.58	0.05	0.03	0.02	0.01	0.02
2 E	0.02	0.05	0.05	0.04	0.03	0.99

Appendix D — Multistart / Solver Evidence

Primary convergence is insufficient: both starts must close and boundary products must agree within 1e-6

N_T	P termination	P closure / residual	S termination	S closure / residual	Branch difference	Gate
1	GTOL	CLOSED / 0.000e+0	GTOL	CLOSED / 0.000e+0	0.000e+0	PASS
2	GTOL	CLOSED / 0.000e+0	GTOL	CLOSED / 0.000e+0	0.000e+0	PASS

MULTISTART INTERPRETATION BOUNDARY

Optimizer termination and numerical residual closure are separate evidence. An unclosed endpoint is not a branch, so branch comparison is not evaluable until both starts close. The equation-closure tolerance remains 1e-8 and the boundary-product agreement tolerance remains 1e-6.

Appendix E — Exact Qualification Blockers

Complete frozen blocker set for every N_T — no truncation

N_T	Residual closure	Exact blocker codes
1	CLOSED	None recorded
2	CLOSED	None recorded

Appendix C.2 — Frozen Global TPD Qualification

Task 216 evidence reconstructed from the closed Project 170 endpoints; never recalculated during PDF generation

Evidence status	BLOCKED_FAIL_CLOSED
Evidence qualified	No
Exact phase coverage	110 / 110
Fully reproduced negative phases	57
Negative or unresolved phases	96
Optimizer/refinement failures	39
Worst frozen TPD minimum	-0.78
Unchanged TPD threshold	-1.00e-8
Task 206 comparison candidate	REJECTED_FROZEN_VALIDATION_AND_GLOBAL_INSTABILITY
Frozen results SHA-256	a8b09497e6d6f087ce417ffcdb6113a59659bfe1de8daa928d923743ce08b5bb
Frozen protocol SHA-256	6876c02629b76d2e564a965b3076390c8e74886ca8eef5d9c67a28b6f8e3b558

Frozen phase classifications

GENUINE_LOWER_GIBBS_BASIN: 57 · LOCAL_HESSIAN_ONLY_ARTIFACT: 14 · OPTIMIZER_REFINEMENT_FAILURE: 39

Exact governing blockers

FROZEN_COMPOSITION_VALIDATION_FAILED · POST_SPLIT_TPD_STABILITY_FAILED · DIRECT_MATCHING_SIX_COMPONENT_LLE_EVIDENCE_MISSING

A completed evidence run with blockers is a controlled negative scientific finding, not a software error. Predictive N_T remains unassigned and the output remains calibration-required and non-release-eligible.

Appendix C.3 — Candidate-Generated Stability Evidence

Immutable candidate cascade lineage; historical Task216/206 comparisons remain diagnostic only

Evidence status	BLOCKED_FAIL_CLOSED
Candidate qualification	BLOCKED — CONTROLLED NEGATIVE
Exact endpoint coverage	110 / 110
Endpoint order	NT ascending, stage ascending, raffinate then extract
Candidate model SHA-256	92700b3b8e24ba63253c33fab7349d605e88101235121a3c5b96cda696860c6b
Candidate parameter SHA-256	a8881b9122e24ef02e80925de1c070f4b025a18e40b31dfe65af64530628cbe5
Candidate flash SHA-256	4c027f09db9a801b85b4b8d4c13ef8c4266c276042910e177c7385ff376a30ba
Cascade / matrix SHA-256	8c634ea01cb580a3f49ba7d57a4951c00cace1632b791201803b4a2dcbb13f20 / 21463f68f2c06ed4d5b9927202bc940e13072c77892bfed6fad2db2fa59ec948
Audit / qualification SHA-256	95bd2f2d8629e03c900ab6d8bce9728101e2624a669b231583cd89fb15f30ffa / b33f00aa1f1b3aca51abc7fd7e5d4634dde8753dd041ce00721b9b4bef10254a

Exact controlling blockers

FROZEN_COMPOSITION_VALIDATION_FAILED • CARRIED_TIE_LINE_GATE_FAILED • DIRECT_MATCHING_SIX_COMPONENT_LLE_EVIDENCE_MISSING • CANDIDATE_CASCADE_LOCAL_OR_GLOBAL_STABILITY_FAILED

Task218 is candidate-generated controlled-negative evidence. It is research-only, calibration-required, not pilot validated, not release eligible; predictive N_T remains unassigned and sulfur remains NOT CALCULABLE.

7. Governance / Report Status

Immutable identifiers for the frozen completed snapshot; full provenance is Appendix F

CONTROLLED RESEARCH OUTPUT — CALIBRATION REQUIRED

Thermodynamic classification	SIX_COMPONENT_COSMO_SAC_2010_RESEARCH_DIAGNOSTIC
Model SHA-256	9666ff67194102ca9c3dc139695bead8ff133fbfe24c2d786ce1deca49456196
Engine SHA-256	2f17fad1c2c68ad408c6ce1543bfc819ed6d903980f3f90469682ae2d9354987
Input SHA-256	—
Stage 1 authority SHA-256	91632843d4ecdfe1661d51fcf385b70f10d0bf1b05c91a911844b0af8b8ceae7
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Job ID	41e150df-e589-4580-98cd-44ecc18d8886

Persisted limitations

- Sulfur removal is NOT CALCULABLE from this aromatic-transfer model.
- The PA representative is non-sulfur-bearing and is not a sulfur surrogate.
- Pilot validation is not established for this persisted snapshot.
- This persisted snapshot is not release eligible.
- Calibration remains required for this persisted snapshot.
- SAT is boundary-mole loss; NMP extraction percentage is not applicable.

DOCUMENT CONTROL

Generated automatically and exclusively from the persisted completed job snapshot. No live design inputs were read and no scientific calculation was rerun.

Appendix F — Governance and Provenance

Immutable identifiers and document-control evidence for the completed snapshot

Thermodynamic classification	SIX_COMPONENT_COSMO_SAC_2010_RESEARCH_DIAGNOSTIC
Model SHA-256	9666ff67194102ca9c3dc139695bead8ff133fbfe24c2d786ce1deca49456196
Engine SHA-256	2f17fad1c2c68ad408c6ce1543bfc819ed6d903980f3f90469682ae2d9354987
Input SHA-256	—
Stage-1 authority SHA-256	91632843d4ecdfe1661d51cf385b70f10d0bf1b05c91a911844b0af8b8ceae7
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Job ID	41e150df-e589-4580-98cd-44ecc18d8886
Completed at	2026-09-01T08:43:29.203Z

DOCUMENT CONTROL

Generated exclusively from the persisted completed job snapshot. The renderer does not read live Stage-1 or workspace values and does not invoke COSMO-SAC, cascade, TPD, or multistart calculations.